

# Yifei YUAN

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## EDUCATION

### The Chinese University of Hong Kong

09/2021 – 07/2026

*Bachelor of Social Science in Economics*

Hong Kong

- Related Courses: Economics Statistics(A), Mathematical Method(A), Machine Learning(A), Economics Research(A)
- Major GPA: 3.81/4 First Class Honours

### University of Geneva

02/2024 – 07/2024

*Semester Exchange Program*

Geneva

- Related Courses: Econometrics(A), Development Economics(A), Game Theory(A), Computational Social Science(A-)
- GPA: 5.8/6 Top 1%

## PUBLICATIONS

- [1] **Yifei Yuan**, Liqun Sun\*, Ji Chen. Solar poverty alleviation program raises local incomes and lowers air pollution in rural China. *Communications Sustainability*. 2026. [\[Paper\]](#) [\[Media\]](#)
- [2] **Yifei Yuan**, Yun Shen, Liqun Sun\*. Next-Generation Application of Perovskite Photovoltaics Unlock Global Solar Potential. (Submitted to *Nature Energy*). 2026

## SELECTED HONOURS & AWARDS

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|------------------------------------------------------------------------------------------------|-------------|
| Full Tuition Scholarship & Honours (Top scorer in Gaokao Admission, Hubei Province)            | 580,000 HKD |
| HKSAR Government Scholarship 2022/23 (One of 838 awardees in Hong Kong)                        | 80,000 HKD  |
| HKSAR Government Scholarship 2023/24 (One of 846 awardees in Hong Kong)                        | 80,000 HKD  |
| Yasumoto International Exchange Scholarship (Covers the full cost of study in Switzerland)     | 20,500 HKD  |
| S.H. Ho Foundation Scholarship (Selected by academic performance and college contributions)    | 10,000 HKD  |
| S.H. Ho College Advanced Scholarship (Top 5 in 450 based on GPA improvement in 2022/23)        | 1,500 HKD   |
| Dean's List Award (Top 10%)                                                                    | Honour      |
| Asia-Pacific Economic Cooperation Scholarship (Awarded by HKSAR with internship opportunities) | Honour      |
| S.H. Ho College Matriculation Scholarships for Academic Excellence (Top 5% at Entrance)        | Honour      |

## RESEARCH INTEREST

My research interests center on the intersection of environmental economics, sustainability assessment, and geospatial technology. I aim to leverage remote sensing applications in social computing, life cycle assessment modeling, and integrated economic-environmental frameworks to evaluate climate policies and emerging carbon neutrality solutions, and to deliver insights for cost-effective and environmentally sound climate action and sustainable development. My current research focuses on the following themes:

- Remote sensing applications in social computing;
- Global-scale assessment of next-generation photovoltaic development;
- Life cycle assessment modeling for sustainability analysis;

## RESEARCH EXPERIENCE

### Chinese Academy of Sciences

Expected: 12/2024 – 08/2026

### Institute of Technology for Carbon Neutrality, SIAT

Shenzhen

*Research Intern (Supervisor: Prof. Sun Liqun)*

- Leveraged ArcGIS and Python to conduct spatiotemporal analysis of global urban nighttime light dynamics, quantifying significant shifts in urban activity patterns across major regions worldwide.
- Participated in the life-cycle assessment and carbon accounting module for a national-level lithium supply chain governance project. Modeled the full carbon footprint across 5 mining-to-battery production facilities in Jiangxi Province, identifying key emission reduction nodes and delivering data-driven recommendations for policy formulation.
- Built a comprehensive panel dataset by integrating multi-source remote sensing and socio-economic data. Implemented

econometric models to identify and quantify the causal effects of environmental policies on regional development.

**The Chinese University of Hong Kong**

09/2025 – 12/2025

**Department of Economics**

Hong Kong

*Dissertation Project (Supervisor: Prof. Pei Guangyu)*

- Built a county-level panel dataset using ArcGIS and Google Earth Engine based on remote sensing data.
- Visualized the energy structure of impoverished counties with Python and constructed PSM-DID models via Stata.
- Empirically verified the significant role of photovoltaic systems in SO<sub>2</sub> emission reduction and revealed the heterogeneous impacts across regions with different income levels and industrial dependence.

**CUHK Business School**

08/2024 – 09/2024

**School of Accountancy**

Hong Kong

*Research Assistant (Supervisor: Prof. Zhou Yuqing)*

- Cleaned and managed economics dataset. Participated in construction of a matched financial dataset of U.S. publicly listed companies and their global subsidiaries, preparing them for analysis and empirical modeling.
- Coordinated with research collaborators and supported research administration tasks, including organizing datasets, literature summaries.

**The Chinese University of Hong Kong**

09/2023 – 12/2023

**Department of Economics**

Hong Kong

*Individual Project (Supervisor: Dr. Vinci Chow)*

- Processed web comment dataset by removing hyperlinks, symbols, and a series of bias factors. Performed word lemmatization and feature engineering to convert nature language to vector.
- Built 5 machine learning classifiers (Logistic Regression, GNB, KNN, Random Forest, SVM), conduct visualization for hyperparameter searching, evaluated model via accuracy/precision/recall/F1, and confirmed SVM as top performers

## PROFESSIONAL EXPERIENCE

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**ByteDance**

09/2023 – 12/2023

*Data Analyst Intern*

Shenzhen

- Conducted research and produced data analysis reports in Southeast Asia market. Provided strategies for operations.
- Participated in e-commerce campaigns and managed weekly operation. Updated strategies based on market report.

**EasyTransfer**

05/2023 – 07/2023

*Global Market Intern*

Beijing

- Conducted market research in Hong Kong and produced 2 reports about the market size of Hong Kong.
- Participated in market expansion activity and conducted community operational data analysis.

## EXTRACURRICULAR ACTIVITIES

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**Ronald McDonald's House Charities**

09/2025 – 05/2026

- Designed and organized weekend retreat workshop series for sick children and their families, incorporating hands-on craft activities and orientation games to boost engagement and provide emotional support.

**Mainland Undergraduate Association Orientation Camp**

08/2022 – 10/2022

- Participated in the design of a series of orientation activities to help new students integrate quickly and collected and analyzed feedback to provide insights for activity experience optimization.

## SKILLS & INTERESTS

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**Language:** English (fluent), Mandarin (native)

**Software:** ArcGIS, STATA, Python, JavaScript, OpenLCA

**Skill:** Global Spatial Analysis, Global Solar Power Simulation, Life Cycle Assessment Modeling, Econometric Modeling